

ALGONA, IA, USA

WMO: 725457

Lat: 43.078N

Lon: 94.272W

Elev: 372

StdP: 96.94

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04904

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.6	-20.1	-27.1	0.3	-22.3	-24.1	0.4	-20.0	14.7	-9.1	13.0	-8.8	4.3	320	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	32.6	24.5	31.2	23.8	29.8	22.9	26.5	30.5	25.3	29.1	24.1	28.1	6.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	21.3	29.2	23.9	19.7	28.2	22.7	18.2	26.8	85.1	30.6	79.6	29.3	74.4	27.8	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.5	11.2	10.0	-27.8	34.7	2.8	1.8	-29.9	36.0	-31.5	37.0	-33.1	38.0	-35.1	39.2		
			-26.6	28.4	3.9	1.3	-29.5	29.3	-31.8	30.0	-34.0	30.7	-36.9	31.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-7.6	-5.3	1.3	8.5	15.5	21.2	22.9	21.4	17.8	10.1	2.4	-4.7	(6)
(7)		DBStd	12.06	6.89	6.86	6.92	6.08	4.95	3.62	3.16	3.07	4.48	5.36	6.20	6.43	(7)
(8)		HDD10.0	2123	544	429	282	98	9	0	0	2	65	238	456		(8)
(9)		HDD18.3	3950	803	662	529	297	117	14	3	64	260	479	713		(9)
(10)		CDD10.0	1646	0	0	12	53	179	336	398	355	235	68	9	0	(10)
(11)		CDD18.3	432	0	0	1	3	28	101	143	104	47	6	0	0	(11)
(12)		CDH23.3	3908	0	0	4	65	292	919	1316	813	428	69	2	0	(12)
(13)		CDH26.7	1246	0	0	1	18	93	321	448	227	118	21	0	0	(13)
(14)	Wind	WSAvg	4.7	5.2	5.3	5.4	5.7	5.4	4.4	3.5	3.2	3.9	4.8	5.1	5.0	(14)
(15)	Precipitation	PrecAvg	774	18	21	46	75	100	118	107	94	81	54	39	23	(15)
(16)		PrecMax	1168	62	88	133	183	204	280	228	232	292	139	143	87	(16)
(17)		PrecMin	437	0	0	2	18	39	33	23	7	14	7	0	0	(17)
(18)		PrecStd	171	13	19	34	41	42	62	52	59	58	36	31	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.0	12.7	22.7	29.0	32.8	34.1	34.1	32.7	32.4	29.6	22.1	13.0	(19)
(20)		MCWB	5.5	10.6	14.7	18.1	21.7	23.2	26.7	25.8	22.5	18.6	14.7	9.6		(20)
(21)		2%	DB	4.8	7.8	17.8	24.9	28.9	32.3	32.5	31.1	30.0	24.8	17.4	8.7	(21)
(22)		MCWB	2.8	5.9	13.1	15.2	19.3	23.2	26.0	25.1	22.0	15.8	12.5	6.8		(22)
(23)		5%	DB	2.5	5.1	14.8	22.0	26.4	30.1	31.1	29.0	27.6	22.1	13.9	5.2	(23)
(24)		MCWB	1.3	3.3	11.2	13.9	17.9	21.8	25.3	23.6	20.6	15.2	9.7	3.2		(24)
(25)		10%	DB	1.2	2.8	11.2	18.1	23.9	28.0	29.1	27.6	26.0	18.8	11.1	2.8	(25)
(26)		MCWB	0.3	1.5	8.6	11.9	16.5	20.7	23.9	22.4	19.4	13.6	7.9	1.5		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	10.4	17.3	19.3	23.1	26.5	28.7	27.4	24.8	20.4	16.0	11.2	(27)
(28)		MADB	8.7	13.0	21.7	26.8	29.3	31.0	32.8	31.3	28.9	26.6	20.3	13.1		(28)
(29)		2%	WB	3.0	6.2	14.0	16.7	20.9	24.7	27.1	25.7	23.5	18.4	12.9	6.3	(29)
(30)		MADB	4.4	7.9	17.0	22.3	27.0	29.6	30.4	29.4	27.8	22.2	16.7	7.9		(30)
(31)		5%	WB	1.4	3.6	11.3	14.8	19.4	23.5	26.0	24.4	22.3	16.1	10.1	3.6	(31)
(32)		MADB	2.4	5.2	14.5	20.6	24.5	28.3	29.8	28.1	26.2	20.2	13.5	5.2		(32)
(33)		10%	WB	0.2	1.8	8.1	12.7	17.9	22.2	24.7	23.2	20.9	13.8	7.7	1.8	(33)
(34)		MADB	0.9	3.0	10.9	17.5	22.2	27.0	28.5	26.7	24.3	18.1	10.5	3.1		(34)
(35)	Mean Daily Temperature Range	MDBR	8.5	8.6	8.9	11.2	11.0	10.8	10.6	10.3	12.1	11.4	9.6	7.9		(35)
(36)		5% DB	MCDBR	9.8	9.9	14.2	17.2	14.2	13.0	11.3	11.4	13.7	15.4	14.4	10.7	(36)
(37)		MCWBR	8.4	8.3	9.5	9.7	7.2	6.2	6.8	6.9	7.3	8.5	9.4	8.7		(37)
(38)		5% WB	MCDBR	9.3	9.6	13.3	15.2	12.4	11.0	10.5	10.3	11.1	12.8	13.6	9.9	(38)
(39)		MCWBR	8.1	8.2	9.4	9.6	7.0	6.8	7.2	7.1	6.8	8.4	9.4	8.3		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.283	0.284	0.312	0.364	0.388	0.412	0.421	0.409	0.373	0.333	0.291	0.268		(40)
(41)		taud	2.406	2.419	2.447	2.330	2.334	2.313	2.264	2.291	2.357	2.462	2.511	2.476		(41)
(42)		Ebn at Noon	858	923	935	901	886	861	848	847	854	851	847	850		(42)
(43)		Edh at Noon	76	89	99	121	124	127	132	124	108	85	69	65		(43)
(44)	All-Sky Solar Radiation	RadAvg	1.85	2.90	3.79	4.71	5.54	6.14	6.45	5.42	4.39	2.93	1.96	1.45		(44)
(45)		RadStd	0.20	0.28	0.38	0.44	0.40	0.48	0.32	0.38	0.29	0.38	0.15	0.12		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page